

INSTITUTE OF ACTUARIES OF INDIA

EXAMINATIONS

7th November 2014

Subject ST1 - Health and Care Insurance

Time allowed: Three hours (14.45* – 18.00 Hrs)

Total Marks: 100

INSTRUCTIONS TO THE CANDIDATES

1. *Please read the instructions on the front page of answer booklet and instructions to examinees sent along with hall ticket carefully and follow without exception*
2. ** You have 15 minutes at the start of the examination in which you are required to read the questions. You are strongly encouraged to use this time for reading only, but notes may be made. You then have three hours to complete the paper.*
3. *You must not start writing your answers in the answer sheet until instructed to do so by the supervisor.*
4. *The answers are not expected to be any country or jurisdiction specific. However, if Examples/illustrations are required for any answer, the country or jurisdiction from which they are drawn should be mentioned.*
5. *Attempt all questions, beginning your answer to each question on a separate sheet.*
6. *Mark allocations are shown in brackets.*
7. *Please check if you have received complete Question Paper and no page is missing. If so, kindly get new set of Question Paper from the Invigilator.*

AT THE END OF THE EXAMINATION

Please return your answer book and this question paper to the supervisor separately.

- Q. 1)** Define the following terms:
- a) Replacement Ratio
 - b) Terminal Illness
 - c) Cost Plus
 - d) Affinity Group
- [4]**
- Q. 2)** A health insurer is proposing to sell its Hospital Cash Plan products through an online web channel.
- Describe the advantages and disadvantages of this proposal.
- [4]**
- Q. 3)** A life insurance company is planning to launch a standalone critical illness benefit product. On diagnosis of one of the four critical illnesses, 100% of the critical illness benefit is paid immediately. A suggestion has been made that:
- i) instead of defining four critical illnesses; it should be defined as any illness, event or condition which renders the policyholder unable to carry out basic bodily function without assistance or
 - ii) to sell the product only by direct marketing and not through tied agents or brokers.
- Discuss the advantage and disadvantage to the company of these proposals.
- [10]**
- Q. 4)** A health and care insurer has issued a large number of group PMI policies in each of the last 5 years.
- i) List the main rating factors for a Group PMI product. (3)
 - ii) Describe how the renewal experience would be analyzed and used in setting re-pricing assumptions. (10)
- The marketing manager has proposed designing two new products on a group business platform; a Group Income protection Insurance Product and a Group Long term care Insurance Product.
- iii) Describe why an employer might purchase a group income protection product for its employees and the features they would expect. (7)
 - iv) Explain why an employer may not be very interested in providing a group long term care insurance product to its employees (2)
- [22]**
- Q. 5)** A health and care insurer has been writing a significant volume of critical insurance business for the last 7 years. In the last 12 months, the new business volume has fallen by 40%. Market analysis suggests that the premium rates of the CI product are almost 25% higher than those of competitors and hence the marketing manager has suggested reducing the premium rates by 25% to regain lost market share.
- i) Describe the factors that should be considered before implementing the suggested premium reduction. (8)

- ii) Describe how the impact of premium rate reduction on the profitability of the product could be determined. (6)
[14]

Q. 6) A health insurance company is experiencing issues with its claims data management system. To overcome these issues, the Chief Actuary, who is also a member of the risk management committee of the company, has proposed developing an in house claims data management system.

- i) Explain why it is important for the company to have a robust claims data management system. (2)
- ii) Describe the areas in which the actuarial department would use information derived from an in house claims data management system. (3)
- iii) Describe the requirements the actuarial department would have of the claims data management system if developed in-house. (3)
- iv) Describe the factors that would be considered in carrying out a technical evaluation of the proposal. (5)
[13]

Q. 7) Describe the factors which influence the determination of interest rate for calculating the statutory reserve for Long Term Health Care plan. [5]

Q. 8) A leading global health insurance company is considering an option to enter into the health insurance market of a new country where it currently does not operate. As a part of its due-diligence exercise, it has approached an actuarial consulting firm for its advice.

Describe the areas in which the actuarial consulting firm could help the insurance company in carrying out this exercise. [10]

Q. 9) There has been an increasing trend in the health insurance market of a developing country to include wellness benefits as an integral component of health insurance product.

- i) Define what is meant by the term wellness benefit. (1)
- ii) Give four examples of a wellness benefit. (2)
- iii) Discuss the advantages of the existence of wellness benefits to health insurance companies writing business in a developing country (3)
[6]

Q. 10) A health insurance company is considering its reinsurance program.

- i) Discuss the factors that would be considered while selecting a reinsurer. (4)

The company's business plan has projected total company's premium of INR 20 crores and an expected claims ratio of 80%.

The expected individual high claims are as follows:

- Claim 1: INR 30 lacs
- Claim 2: INR 50 lacs
- Claim 3: INR 1.10 crores
- Claim 4: INR 1.50 crores
- Claim 5: INR 1.70 crores

The company has INR 30 crores of capital including regulatory capital. Regulatory capital is calculated as 10% of premium after reinsurance.

- ii)** The company is concerned about the effect of unfavorable large claims volatility on its capital and profitability. It has received two quotes from two different reinsurers.

Quote 1: 50/50 Quota Share on original premium basis after allowing for expense loading of 10% on original premium.

Quote 2: Reinsurer takes 75% of risk for an individual claim in excess of INR 25 lacs but up to INR 1.00 crore. The total reinsurance premium for the year is INR 2.50 crores for the portfolio.

- a)** Calculate the effects of the two proposed reinsurance quotes on the insurance company's profits and regulatory capital using assumptions given above. (4)
- b)** Discuss the advantage and disadvantage of each of the reinsurance quotes. (4)

[12]
